



CONTACT

📍 Isfahan, Iran
📞 +98 914 040 9147
✉ mohammadsoroushrabiei@gmail.com
🐙 github.com/MohammadSoroushRabiei
🌐 linkedin.com/in/mohammad-soroush-rabiei-0138a6368

TECHNICAL SKILLS

Programming

C Embedded C C++ Python

MCUs

STM32 (F4/F7) ESP32 / ESP32-S3
ESP8266

Frameworks & Tools

STM32CubeMX/IDE Keil MDK ESP-IDF
Git Doxygen

Protocols & Peripherals

UART I²C RS-485 Modbus RTU
MQTT DMA Timer input capture

Debug & Test

ST-LINK Logic analyzer
Hardware bring-up Simulators

RTOS

FreeRTOS (tasks, queues, watchdogs)

AI / ML

PyTorch (coursework)
OpenCV (in progress) ESP-DL

LANGUAGES

🇮🇷 Persian native
🇬🇧 English upper-intermediate (B2)
🇨🇳 Chinese elementary (≈HSK 2)

Mohammad Soroush Rabiei

Embedded Systems Developer

SUMMARY

Embedded systems developer with industrial experience in data-center infrastructure management (DCIM). My work and interests span IoT and connected devices, edge AI, and building smart products — mostly at the boundary of hardware and software.

EXPERIENCE

- **Embedded Systems Developer**
Kian Pardaz Nagh-e Jahan · Isfahan, Iran
Jul 2026 – Present
 - ▶ Researched and selected temperature/humidity sensors for the DCIM product.
 - ▶ Developed production STM32 drivers — a non-blocking DHT22 driver (timer input-capture + DMA) and a Modbus RTU master over RS-485 — open-sourced with full documentation.
- **Embedded Systems Intern**
Behbood Ertebat Chehelsotoon Co. (Behamad) · Isfahan, Iran
Jun 2025 – Sep 2025
 - ▶ Developed STM32 firmware for a smart-city (municipal IoT) device with LoRaWAN and a SIM800 GSM module.
 - ▶ Implemented MQTT on STM32 to connect a smart thermostat to the platform.

PROJECTS

Smart Home — Self-Hosted Edge-AI Platform · B.Sc. final project

- 🐙 github.com/MohammadSoroushRabiei/Smart-Home
- ▶ A cloud-free smart home on ESP32-S3: on-device face recognition (ESP-DL), a self-learning on-chip behavior agent, an LVGL touch dashboard and a fully local MQTT/Home Assistant stack.

IoT Smart Room · IoT course project

- 🐙 github.com/MohammadSoroushRabiei/IoT_SmartRoom_ESP8266
- ▶ Remote room control and monitoring (lights, fan, temperature) on ESP8266 with Blynk.

EDUCATION

- **B.Sc. in Computer Engineering**
Intelligent Systems · Isfahan University of Technology
2020 – 2026
- **Diploma, Mathematics & Physics**
NODET (SAMPAD) High School, Isfahan
2018